Panda kinematics: where every number comes from

Challenge 01 | code: panda_kinematics/kinematics.py | model: Challenge 01/model/panda.xml | all angles in radians (`<compiler angle="radian">`), all lengths in meters

1. Constants copied from panda.xml (and the two that are not)

In kinematics.py	Value in our code	Exact source in Challenge 01/model/panda.xml	XML line

Q_MIN / Q_MAX
joints 1, 3, 5, 7
	`-2.8973 .. 2.8973`	`<default class="panda">`
		`<joint ... range="-2.8973 2.8973"/>`
		`(joint1/3/5/7 have no range of their own, so they inherit it)`
			9
			(138, 148, 168, 200)

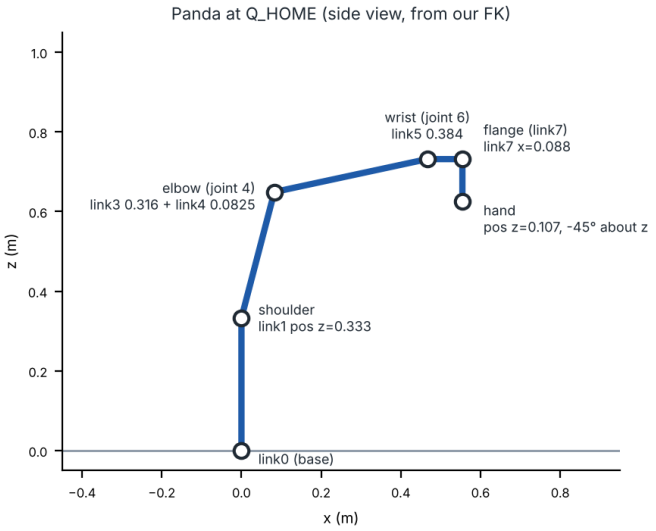
Q_MIN / Q_MAX joint 2	`-1.7628 .. 1.7628`	`<joint name="joint2" range="-1.7628 1.7628"/>`	144
Q_MIN / Q_MAX joint 4	`-3.0718 .. -0.0698`	`<joint name="joint4" range="-3.0718 -0.0698"/>`	159
Q_MIN / Q_MAX joint 6	`-0.0175 .. 3.7525`	`<joint name="joint6" range="-0.0175 3.7525"/>`	178
T_L7_HAND translation	`0.107 (last column, z)`	`<body name="hand" pos="0 0 0.107" ...>`	210
T_L7_HAND rotation	`cos(-pi/4), sin(-pi/4)` `= -45 deg about z`	`quat="0.9238795 0 0 -0.3826834"` `= (cos(t/2), 0, 0, sin(t/2)), t/2 = -22.5 deg`	210
Q_HOME	`[0, 0, 0, -1.57079, 0,` `1.57079, -0.7853]`	`<key name="home" qpos="0 0 0 -1.57079 0 1.57079 -0.7853 0.04 0.04"/>` `(first 7 = arm joints; last 2 = fingers, not used)`	281
SHOULDER	`[0, 0, 0.333]`	`<body name="link1" pos="0 0 0.333">`	135
TARGET_JOINTS	`[0.5, 0.3, -0.2, -2.0,` `0.1, 2.3, 0.4]`	**Not from the XML.** Picked by hand: every value is inside Q_MIN/Q_MAX and puts the hand in front of the robot, pointing down.	-
TARGET_POSE	`forward_kinematics(` `TARGET_JOINTS)`	**Not from the XML.** Computed by our FK, so the target is reachable by construction. It is `None` at first only because the function is defined later in the file.	-

Flow: TARGET_JOINTS (7 angles) → forward_kinematics → TARGET_POSE (4x4: position + orientation) → inverse_kinematics → joint angles sent to the robot. **range** = where a joint can physically move; **ctrlrange** = the command values its motor accepts (equal for the arm joints, 0-255 for the gripper).

2. DH parameters [alpha, a, d]

Convention: A_i = Rz(theta_i) . Tz(d_i) . Tx(a_i) . Rx(alpha_i). theta_i is the joint angle q_i, so it is not in the table. MuJoCo places each child body with T(pos) . R(quat); quat "1 -1 0 0" = Rx(-90 deg), "1 1 0 0" = Rx(+90 deg).

i	alpha	a (m)	d (m)	Where it comes from in panda.xml (line)
1	-pi/2	0	0.333	link1 pos z = 0.333 (135); link2 quat "1 -1 0 0" = Rx(-90) (141)
2	+pi/2	0	0	link3 quat "1 1 0 0" = Rx(+90) (147)
3	+pi/2	0.0825	0.316	link3 pos y = -0.316 becomes +z after Rx(+90) (147); link4 pos x = 0.0825, quat Rx(+90) (156)
4	-pi/2	-0.0825	0	link5 pos x = -0.0825, quat "1 -1 0 0" = Rx(-90) (165)
5	+pi/2	0	0.384	link5 pos y = 0.384 becomes +z after Rx(-90) (165); link6 quat Rx(+90) (175)
6	+pi/2	0.088	0	link7 pos x = 0.088, quat Rx(+90) (197)
7	0	0	0	frame 7 = link7 body; the 0.107 to the hand is in T_L7_HAND (210)



3. Reading the angles in degrees

Array	Radians (as in code)	Degrees (rad x 180 / pi)
Q_HOME	`0, 0, 0, -1.57079, 0, 1.57079, -0.7853`	`0.0, 0.0, 0.0, -90.0, 0.0, 90.0, -45.0`
TARGET_JOINTS	`0.5, 0.3, -0.2, -2, 0.1, 2.3, 0.4`	`28.6, 17.2, -11.5, -114.6, 5.7, 131.8, 22.9`

4. TARGET_POSE = forward_kinematics(TARGET_JOINTS)

`0.8092`	`0.5875`	`0.0029`	`0.5730`
`0.5873`	`-0.8092`	`0.0175`	`0.1837`
`0.0126`	`-0.0124`	`-0.9998`	`0.3126`
`0.0000`	`0.0000`	`0.0000`	`1.0000`

Blue 3x3 = orientation of the hand.
Orange column = position x, y, z in meters (0.573, 0.184, 0.313).
Third column ~ (0, 0, -1): the hand's z axis points down.

Checked against MuJoCo itself. panda.xml was loaded in MuJoCo 3 and the 'hand' body pose was compared with `forward_kinematics(q)` for Q_HOME, TARGET_JOINTS and 200 random joint vectors inside the limits: largest difference 3.5e-8 m (0.00003 mm). MuJoCo's joint ranges and home keyframe equal Q_MIN, Q_MAX and Q_HOME exactly. IK from Q_HOME reaches TARGET_POSE (converged = True).